



ARTIFICIAL INTELLIGENCE

Class XI

STUDENT HANDBOOK



Subject Code:843

UNIT 6: Machine Learning Algorithms

Title: Machine Learning Algorithms	Approach: Interactive/ Collaborative / Hands on Activity
Summary: In this unit, the students will be able to understand the concept of Machine Learning and its connection with AI. They will understand the different ways in which machines can learn or can be taught using different methodologies. The main types of machine learning methods- Supervised, Unsupervised and Reinforcement methods will be introduced to the students.	
Learning Objectives: 1. Understand Machine Learning and the various machine learning algorithms 2. Understand regression as a type of supervised learning. 3. Understand classification as a type of supervised learning. 4. Understand clustering as a type of unsupervised learning. 5. List of algorithms for regression, classification and clustering 6. Differentiate between regression problem, classification problem and clustering problem.	
Key Concepts: 1. Machine Learning in a nutshell 2. Types of Machine Learning 3. Supervised Learning - Regression - Understanding Correlation, Regression, Finding the line, Linear Regression algorithm - Classification – How it works, Types, k – Nearest Neighbour algorithm 4. Unsupervised Learning - Clustering – How it works, Types, k -means Clustering algorithm	
Learning Outcomes: By the end of this unit, students will be able to: 1. Differentiate the different types of machine learning methods. 2. They will be able to understand the concept behind each machine learning methods. 3. Apply these methods to develop simple solutions for some day-to-day situations. 4. Build up this knowledge to the next level to apply during Capstone Project development.	
Pre-requisites: Basic Knowledge about AI, domains of AI, applications of AI.	

Students, you have likely experienced the following scenarios in your daily lives:

- When browsing through online shopping apps, you receive recommendations to purchase socks immediately after looking at shoes.
- Online entertainment platforms suggest movies similar to ones you have previously watched.
- Facial recognition technology identifies you in photos on social media platforms.
- Chatbots engage in conversations with you in a natural manner.

MACHINE LEARNING IN A NUTSHELL

Machine Learning (ML) is a part of artificial intelligence (AI) that focuses on teaching computers to learn from data and make decisions without being explicitly programmed. Unlike traditional programming where developers provide precise instructions, ML algorithms learn from patterns and relationships in data. This allows them to generalize and make decisions on new, unseen data.

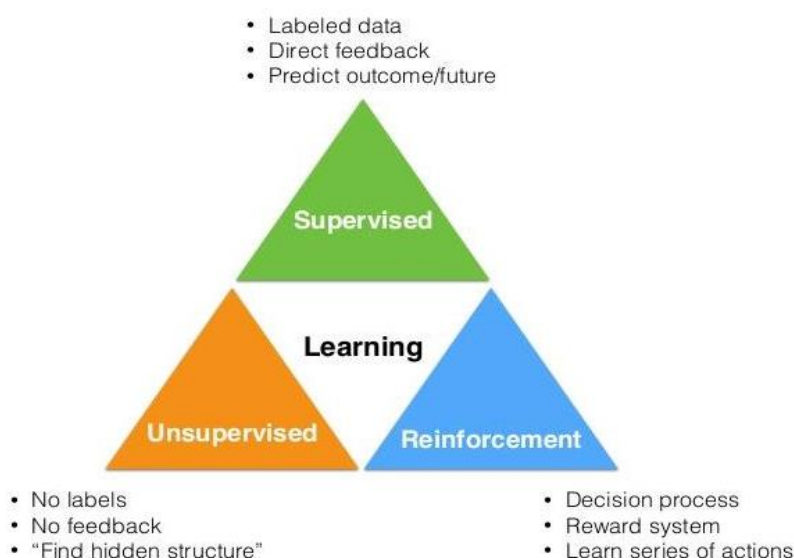
- ML algorithms learn from various **types of data**, including images, text, sensor readings, and historical records. Instead of hardcoding rules, ML models identify patterns and relationships within the data to make predictions or decisions.
- Some common ML **algorithms** include decision trees, neural networks, and support vector machines. Trained models serve as representations of the learned data, such as recognizing handwritten digits using a neural network.
- The **applications** of ML are vast and diverse. It powers recommendation systems like those used by Netflix, speech recognition, medical diagnosis, and autonomous vehicles. ML is also behind chatbots, personalized ads, and fraud detection systems.
- However, ML also presents **challenges**. Overfitting, where models become too specialized on training data, can lead to poor performance on new data. Bias in training data can result in biased predictions, and some models are difficult to interpret, acting as black boxes. Despite these challenges, ML transforms data into knowledge, enabling computers to learn, adapt, and make decisions autonomously.
- Artificial intelligence (AI) and machine learning (ML) have significantly **impacted** various aspects of our lives. From transportation and finance to healthcare and entertainment, AI algorithms are pervasive. They power self-driving cars, fraud detection systems, personalized shopping experiences, and virtual assistants like Siri and Alexa. As technology continues to evolve, the influence of AI and ML is only expected to grow, shaping the future of our society and culture.

Activity 1: [Autodraw](#) - Experience the power of machine learning with Autodraw! Autodraw combines machine learning with the creativity of talented artists, allowing you to draw things quickly and effortlessly.

TYPES OF MACHINE LEARNING

In Chapter 1, we introduced you to the fascinating world of artificial intelligence (AI) and its various learning mechanisms. We discussed three main types of machine learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. These terms represent the algorithms that drive AI systems, serving as the building blocks for programming intelligent behavior and decision-making processes. Now, let us delve deeper into how these algorithms shape the landscape of AI applications.

- **Supervised learning** involves the model learning from **labeled data**, where the input data is accompanied by the correct output. The algorithm learns to **map input data to output labels** based on example input-output pairs provided during training. The goal is to learn a mapping function so that the model can make predictions on unseen data. Examples include *linear regression, logistic regression, decision trees, support vector machines, and neural networks*.
- **Unsupervised learning**, on the other hand, deals with **unlabelled data**, where the algorithm tries to **find hidden patterns or structure** without explicit guidance. The goal of this is to explore and discover inherent structures or relationships within the data, such as clusters or associations. Examples include *k-means clustering, hierarchical clustering, principal component analysis, and autoencoders*.
- Finally, **reinforcement learning** involves an agent learning to make decisions by interacting with an environment to maximize cumulative rewards. Through **trial and error**, the agent learns a policy or strategy to take actions that lead to the highest cumulative reward over time. These **rewards** serve as feedback, guiding the agent towards favorable actions, while **penalties** discourage undesirable behavior. This iterative learning approach is particularly effective in scenarios requiring a sequence of decisions, such as game-playing or robot control, where the agent adapts its behavior based on past experiences and outcomes. Examples include *Q-learning, deep Q-networks, policy gradients, and actor-critic methods*.



A. SUPERVISED LEARNING

Supervised learning stands out as one of the foundational types of Machine Learning. It is a powerful approach that allows machines to learn from labeled data, making predictions or decisions based on that learning. Within supervised learning, two primary types of algorithms emerge:

1. Regression – works with continuous data
2. Classification – works with discrete data

1. REGRESSION

Understanding Correlation: The Foundation of Regression Analysis

In data analysis, correlation is a fundamental concept that helps us grasp the relationship between variables, laying the groundwork for predictive modeling and insightful analysis. Correlation is a measure of the strength of a linear relationship between two quantitative variables (e.g. price, sales). If the change in one variable appears to be accompanied by a change in the other variable the two variables are said to be **correlated** and this inter dependence is called **correlation**.

Types of Correlation:

1. Positive Correlation: In a positive correlation, both variables move in the same direction. As one variable increases, the other also tends to increase, and vice versa.	Salary -> Dependent Variable Experience -> Independent Variable Perfect Positive Correlation High Positive Correlation Low Positive Correlation 1 0.9 0.5 Experience Increase then Salary Increase
2. Negative Correlation: Conversely, in a negative correlation, variables move in opposite directions. An increase in one variable is associated with a decrease in the other, and vice versa.	Low Negative Correlation High Negative Correlation Perfect Negative Correlation -0.5 -0.9 -1 Vehicle: Age Increase Value Decrease
3. Zero Correlation: When there is no apparent relationship between two variables, they are said to have zero correlation. Changes in one variable do not predict changes in the other	No Correlation 0

Correlation can have a value:

- 1 is a perfect positive correlation
- 0 is no correlation (the values don't seem linked at all)
- -1 is a perfect negative correlation

Causation

Causation indicates that one event is the result of the occurrence of the other event. Example: Since there is hot weather, the person will use more sunscreen or eat more ice cream.

Sometimes, these two events may be correlated also. Example: smoking causes an increase in the risk of developing lung cancer or it can correlate with another like-smoking is correlated with alcoholism, but it does not cause alcoholism. Therefore, we can say causation is not always correlation.

PEARSON'S R

Pearson's correlation coefficient (often denoted as Pearson's **r**) is one of the crucial factors to consider when assessing the appropriateness of regression analysis. **Pearson's r measures the strength and direction of the linear relationship between two continuous variables.** In the context of regression analysis, a high degree of correlation between the independent and dependent variables suggests that there may be a meaningful relationship to explore using regression techniques.

The **requirements** when considering the use of Pearson's correlation coefficient are:

1. Scale of measurement should be interval or ratio.
2. Variables should be approximately normally distributed.
3. The association should be linear.
4. There should be no outliers in the data.

Pearson's **r** is calculated using the formula:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

r can take a range of values from +1 to -1

- A value of 0 indicates that there is no association between the two variables.
- A value greater than 0 indicates a positive association; that is, as the value of one variable increases, so does the value of the other variable.
- A value less than 0 indicates a negative association; that is, as the value of one variable increases, the value of the other variable decreases.

Example 1

In the example below of 6 people with different ages and different weight, let us try calculating the value of the Pearson **r**.

Sr. No	Age (x)	Weight (y)
1	40	78
2	21	70
3	25	60
4	31	55
5	38	80
6	47	66

Solution:

For the Calculation of the Pearson Correlation Coefficient, we will first calculate the following values:

Sr. No	Age (x)	Weight (y)	xy	x ²	y ²
1	40	78	3120	1600	6084
2	21	70	1470	441	4900
3	25	60	1500	625	3600
4	31	55	1705	961	3025
5	38	80	3040	1444	6400
6	47	66	3102	2209	4356
Total (Σ)	202	409	13937	7280	28365

Here the total number of people is 6 so, **n=6**

Now the calculation of the Pearson R is as follows:

E12	$=(6*D10-B10*C10)/SQRT((6*E10-B10^2)*(6*F10-C10^2))$				
	A	D	E	F	G
3	Sr. No	Age (x)	Weight (y)	xy	x ²
4	1	40	78	3120	1600
5	2	21	70	1470	441
6	3	25	60	1500	625
7	4	31	55	1705	961
8	5	38	80	3040	1444
9	6	47	66	3102	2209
10	Total (Σ)	202	409	13937	7280
11					
12	Pearson Correlation Coefficient (r)	0.35			

$$r = (n (\sum xy) - (\sum x)(\sum y)) / (\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]})$$

$$r = (6 * (13937) - (202)(409)) / (\sqrt{[6 * 7280 - (202)^2] * [6 * 28365 - (409)^2]})$$

$$r = (6 * (13937) - (202) * (409)) / (\sqrt{[6 * 7280 - (202)^2] * [6 * 28365 - (409)^2]})$$

$$r = (83622 - 82618) / (\sqrt{[43680 - 40804] * [170190 - 167281]})$$

$$r = 1004 / (\sqrt{[2876] * [2909]})$$

$$r = 1004 / (\sqrt{8366284})$$

$$r = 1004 / 2892.452938$$

$$r = 0.35$$

The value of the Pearson correlation coefficient is **0.35**

It is important to note that, regression analysis may not be suitable in certain situations:

1. **No Correlation:** If there is no correlation between the variables, meaning they change independently of each other, regression analysis will not provide meaningful insights or predictions.
2. **Non-linear Relationships:** While regression can model linear relationships well, it may not capture more complex, non-linear relationships effectively. In such cases, alternative techniques like polynomial regression or non-linear regression may be more appropriate.

3. **Outliers:** Outliers, or extreme data points, can disproportionately influence the regression model and lead to inaccurate predictions. In the presence of outliers, it is essential to assess their impact and consider alternative modeling approaches.
4. **Violation of Assumptions:** Regression analysis relies on certain assumptions, such as the linearity of relationships and the absence of multicollinearity (high correlation between predictor variables). If these assumptions are violated, the results of the regression analysis may be unreliable.

REGRESSION

Regression is a statistical technique used to model the relationship between a dependent variable and one or more independent variables. Its primary objective is to understand and predict the value of the dependent variable based on the values of the independent variables. In simpler terms, regression helps us understand how changes in one or more variables are associated with changes in another variable.

Regression analysis is particularly useful when dealing with **continuous data**, where variables can take on any value within a certain range. For example, variables such as height, temperature, salary, and time are all continuous, meaning they can be measured along a continuous scale. In regression, these continuous variables are used to predict or explain the variability in another continuous variable, known as the dependent variable. By analyzing the relationship between the independent and dependent variables, regression allows us to make predictions and understand how changes in one variable may impact the other. This makes regression a powerful tool for forecasting, prediction, and understanding complex relationships in various fields such as economics, social sciences, and healthcare.

When we make a distribution in which there is an involvement of more than one variable, then such an analysis is called Regression Analysis. It generally focuses on finding or rather predicting the value of the variable that is dependent on the other.

Let there be two variables x and y . If y depends on x , then the result comes in the form of a simple regression. Furthermore, we name the variables x and y as:

y – Regression / Dependent / Explained Variable.
It is the variable we want to predict or understand.

x – Independent / Predictor / Explanator Variable
It is used to predict or explain changes in the dependent variable.

Therefore, if we use a simple linear regression model where y depends on x , then the regression line of y on x is:

$$y = a + bx + e$$

In this equation,

- **a** represents the **intercept** of the regression line with the y -axis.
- **b** represents the **slope** of the regression line, indicating the rate of change in y for a unit change in x .
- **e** represents the **error or residual**, which accounts for the difference between the observed values of y and the values predicted by the regression equation.

FINDING THE LINE

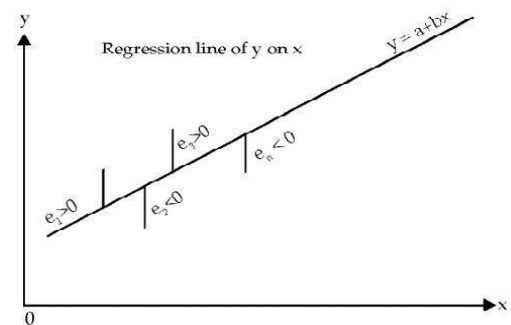
Regression analysis relies on the concept of the regression line or curve, which represents the best-fit relationship between the variables involved. This line or curve is determined by minimizing the differences between the observed values of the dependent variable and the values predicted by the regression model.

The **least squares method** is commonly employed to find this best-fit line or curve. This method minimizes the squared differences between observed and predicted values, ensuring that the regression line captures the overall trend or pattern in the data as accurately as possible. By systematically estimating the parameters of the regression model, such as the intercept and slope, the least squares method provides a precise description of the relationship between the variables.

Through the least squares method, regression analysis yields estimate of the regression coefficients that define the best-fit relationship between the variables. These coefficients allow for making predictions about the dependent variable based on the values of the independent variable(s) with greater accuracy and reliability. As a result, this is widely used in regression analysis.

Properties of the Regression line:

- The line minimizes the sum of squared difference between the observed values (actual y-value) and the predicted value ($\hat{y}$ value)
- The line passes through the mean of independent and dependent features.



Example 1

In the example of 6 people with different ages and different weight, let us draw the line of best fit in Excel.

Sr. No	Age (x)	Weight (y)
1	40	78
2	21	70
3	25	60
4	31	55
5	38	80
6	47	66

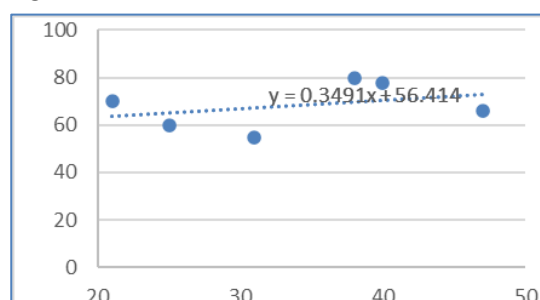
Solution:

Step 1: Select the Age and Weight.

Step 2: Insert a scatter chart and make changes to the following:

Trendline Name: **Linear**, check **Display Equation on Chart**

X axis **minimum : 20**



Step 3: Let us verify the values of slope and intercept using slope() and intercept() function in excel.

Step 4: Click on any cell and type =slope(Now, select the values of Weight, and then type comma. Now, select the values of Age and press enter.

Step 5: Click on any cell and type =intercept(Now, select the values of Weight, and then type comma. Now, select the values of Age and press enter.

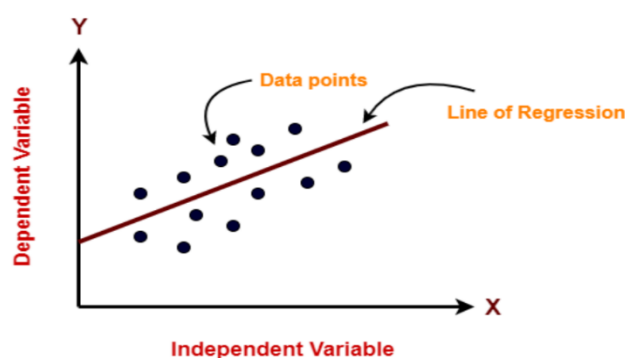
Some of the regression algorithms include Linear Regression, Logistic Regression, Decision Tree Regression, Random Forest Regression. Let us learn about Linear Regression.

Linear Regression

Linear regression is one of the most basic types of regression in machine learning. The linear regression model consists of a predictor variable and a dependent variable related linearly to each other. In case the data involves more than one independent variable, then linear regression is called multiple linear regression models.

Linear regression is further divided into two types:

- a) **Simple Linear Regression:** The dependent variable's value is predicted using a single independent variable in simple linear regression.
- b) **Multiple Linear Regression:** In multiple linear regression, more than one independent variable is used to predict the value of the dependent variable.



Applications of Linear Regression:

- **Market Analysis:** Linear regression helps understand how different factors like pricing, sales quantity, advertising, and social media engagement relate to each other in the market.
- **Sales Forecasting:** It predicts future sales by analyzing past sales data along with factors like marketing spending, seasonal trends, and consumer behavior.
- **Predicting Salary Based on Experience:** Linear regression estimates a person's salary based on their years of experience, education, and job role, aiding in recruitment and compensation planning.
- **Sports Analysis:** Linear regression analyzes player and team performance by considering statistics, game conditions, and opponent strength, assisting coaches and team management in decision-making.
- **Medical Research:** Linear regression examines relationships between factors like age, weight, and health outcomes, helping researchers identify risk factors and evaluate interventions.

Advantages of Linear regression

- Simple technique and easy to implement
- Efficient to train the machine on this model

Disadvantages of Linear regression

1. Sensitivity to outliers, which can significantly impact the analysis.
2. Limited to linear relationships between variables.

<https://www.javatpoint.com/linear-regression-in-machine-learning>

For Advanced Learners – Python program for Linear regression

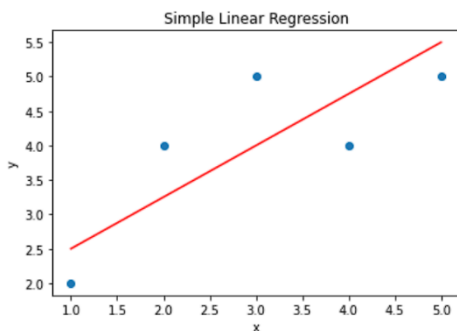
Import **scipy** and draw the line of Linear Regression:

```
import numpy as np
import matplotlib.pyplot as plt
# Sample data
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 4, 5])
# Calculate mean and standard deviation
x_mean = np.mean(x)
y_mean = np.mean(y)
x_std = np.std(x)
y_std = np.std(y)
# Calculate covariance and slope
covariance = np.sum((x - x_mean) * (y - y_mean)) / (len(x) - 1)
slope = covariance / (x_std**2)
# Calculate y-intercept (b)
intercept = y_mean - slope * x_mean
# Predicted values
y_pred = slope * x + intercept
# Plot data and regression line
plt.scatter(x, y)
plt.plot(x, y_pred, color='red')
# Add labels and title
plt.xlabel('x')
plt.ylabel('y')
plt.title('Simple Linear Regression')
# Show the plot
plt.show()
# Print slope and intercept
print(f"Slope: {slope:.2f}")
print(f"Intercept: {intercept:.2f}")
```

This program:

- Imports numpy for numerical calculations and matplotlib.pyplot for plotting.
- Defines sample data for x and y. You can replace this with your own data.
- Calculates mean, standard deviation, covariance, and slope.
- Calculates y-intercept based on slope and mean.
- Predicts y values for given x using the linear equation.
- Plots the data points and the regression line.
- Prints the estimated slope and intercept values.

The expected output of the above program would be



Slope: 0.75
Intercept: 1.75

REFERENCES

Video links:

- <https://www.youtube.com/watch?v=rHeaoaiBM6Y>
- https://www.youtube.com/watch?v=VWCRDH1_rv0
- https://www.youtube.com/watch?v=YUPagM-OB_M
- <https://www.youtube.com/watch?v=CtsRRUddV2s&t=70s>

2. CLASSIFICATION

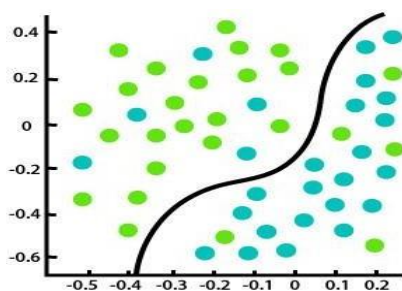
Classification is a fundamental concept in artificial intelligence and machine learning that involves categorizing data into predefined classes or categories. The main objective of classification is to assign labels to data instances based on their features or attributes. In classification, the data is typically labeled with class labels or categories, and the goal is to build a model that can accurately assign these labels to new, unseen data instances. This process is supervised learning, where the model learns from labeled training data to make predictions on unseen data.

For example, let us say, you live in a gated housing society and your society has separate dustbins for different types of waste: paper waste, plastic waste, food waste and so on. What you are basically doing over here is classifying the waste into different categories and then labeling each category. In the picture given below, we are assigning the labels 'paper', 'metal', 'plastic', and so on to different types of waste.

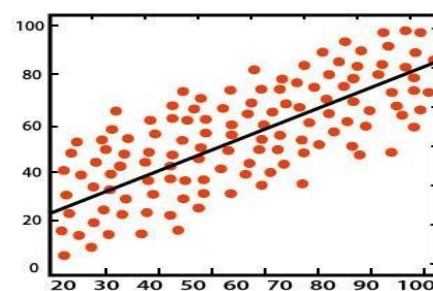


Look at the two graphs below and suggest which graph represents the classification problem.

Graph 1



Graph 2



How Classification Works

In classification tasks within machine learning, the process revolves around categorizing data into distinct groups or classes based on their features. Here is how it typically works:

- **Classes or Categories:** Data is divided into different classes or categories, each representing a specific outcome or group. For example, in a binary classification scenario, there are two classes: positive and negative.
- **Features or Attributes:** Each data instance is described by its features or attributes, which provide information about the instance. These features are crucial for the classification model to differentiate between different classes. For instance, in email classification, features might include words in the email text, sender information, and email subject.
- **Training Data:** The classification model is trained using a dataset known as training data. This dataset consists of labelled examples, where each data instance is associated with a class label. The model learns from this data to understand the relationship between the features and the corresponding class labels.

- **Classification Model:** An algorithm or technique is used to build the classification model. This model learns from the training data to predict the class labels of new, unseen data instances. It aims to generalize from the patterns and relationships in the training data to make accurate predictions.
- **Prediction or Inference:** Once trained, the classification model is used to predict the class labels of new data instances. This process, known as prediction or inference, relies on the learned patterns and relationships from the training data..

Types of classification

The four main types of classification are:

- 1) Binary Classification
- 2) Multi-Class Classification
- 3) Multi-Label Classification
- 4) Imbalanced Classification

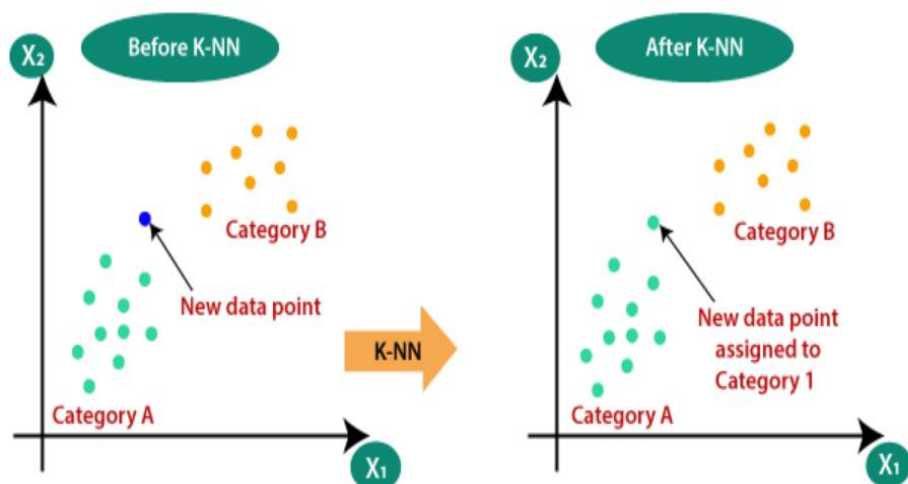
Classification Type	Binary Classification	Multi-Class Classification	Multi-Label Classification	Imbalanced Classification
Description	Classification tasks with two class labels.	Classification tasks with more than two class labels.	Classification tasks where each example may belong to multiple class labels.	Classification tasks with unequally distributed class labels, typically with a majority and minority class.
Example	• Email spam detection - spam or not • Conversion prediction - buy or not • Medical test - Cancer detected or not • Exam results - pass/fail	• Face classification • Plant species classification • Optical character recognition • Image classification into thousands of classes	• Photo classification - objects present in the photo (bicycle, apple, person, etc.)	• Fraud detection • Outlier detection • Medical diagnostic tests

K- Nearest Neighbour algorithm (KNN)

The K-Nearest Neighbors algorithm, commonly known as KNN or k-NN, is a versatile non-parametric supervised learning technique used for both classification and regression tasks. It operates based on the principle of proximity, making predictions or classifications by considering the similarity between data points.

Why KNN Algorithm is Needed:

KNN is particularly useful when dealing with classification problems where the decision boundaries are not clearly defined or when the dataset does not have a well-defined structure. It provides a simple yet effective method for identifying the category or class of a new data point based on its similarity to existing data points.



Steps involved in k-NN

- Select the number K of the neighbors
- Calculate the Euclidean distance of K number of neighbors
- Take the K nearest neighbors as per the calculated Euclidean distance.
- Among these k neighbors, count the number of the data points in each category.
- Assign the new data points to that category for which the number of the neighbor is maximum.
- Our model is ready.

Applications of KNN:

- Image recognition and classification
- Recommendation systems
- Healthcare diagnostics
- Text mining and sentiment analysis
- Anomaly detection

Advantages of KNN:

- Easy to implement and understand.
- No explicit training phase; the model learns directly from the training data.
- Suitable for both classification and regression tasks.
- Robust to outliers and noisy data.

Limitations of KNN:

- Computationally expensive, especially for large datasets.
- Sensitivity to the choice of distance metric and the number of neighbors (K).
- Requires careful preprocessing and feature scaling.
- Not suitable for high-dimensional data due to the curse of dimensionality.

For advanced learners – Python Program for K Nearest Neighbor Algorithm

importing libraries

import numpy as nm

import matplotlib.pyplot as mtp

import pandas as pd

#importing datasets

data_set= pd.read_csv('user_data.csv')

#Extracting Independent and dependent Variable

x= data_set.iloc[:, [2,3]].values

y= data_set.iloc[:, 4].values

Splitting the dataset into training and test set.

from sklearn.model_selection **import** train_test_split

x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=0)

#feature Scaling

from sklearn.preprocessing **import** StandardScaler

st_x= StandardScaler()

x_train= st_x.fit_transform(x_train)

x_test= st_x.transform(x_test)

Reference: <https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning>

Note to the teacher: Can introduce Teachable Machine to develop any image classifier problem.

Website: <https://teachablemachine.withgoogle.com/>

REFERENCES

Video Session:

Classification: <https://www.youtube.com/watch?v=hBKI7XvD8R8>

KNN Algorithm: <https://www.youtube.com/watch?v=0p0o5cmgLdE>

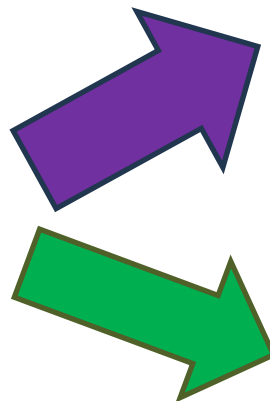
B. UNSUPERVISED LEARNING

3. CLUSTERING

Clustering, or cluster analysis, is a machine learning technique used to group **unlabeled dataset** into clusters or groups based on similarity. Clustering aims to organize data points into groups where points within the same group are more similar to each other than to those in other groups. It involves finding patterns or structures in the data without the need for predefined class labels. It does it by finding some similar patterns in the unlabelled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns. It is an **unsupervised learning** method, hence no supervision is provided to the algorithm, and it deals with the unlabeled dataset.

The clustering technique is commonly used for **statistical data analysis**.

Example: Let us consider the clustering technique using a real-world example. Imagine you are visiting a shopping center where items are grouped together based on their similarities. For instance, in the fruits section, you will find apples, bananas, and grapes neatly arranged together. This organization makes it convenient for shoppers to locate specific items they are looking for.



Based on colour



Based on size

In a similar way, clustering algorithms group similar data points together based on common characteristics or features. This approach helps in organizing and making sense of large datasets in various tasks, such as market segmentation, image recognition, and customer segmentation.

How Clustering Works:

To cluster data effectively, follow these key steps:

- 1) **Prepare the Data:** Select the right features for clustering and make sure the data is ready by scaling or transforming it as needed.
- 2) **Create Similarity Metrics:** Define how similar data points are by comparing their features. This similarity measure is crucial for clustering.
- 3) **Run the Clustering Algorithm:** Apply a clustering algorithm to group the data. Choose one that works well with your dataset size and characteristics.
- 4) **Interpret the Results:** Analyze the clusters to understand what they represent. Since clustering is unsupervised, interpretation is essential for assessing the quality of the clusters.

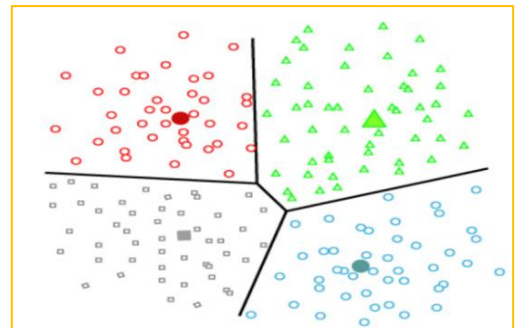
Types of Clustering Methods

Some of the common clustering methods used in Machine learning are:

- 1) Partitioning Clustering
- 2) Density-Based Clustering
- 3) Distribution Model-Based Clustering
- 4) Hierarchical Clustering

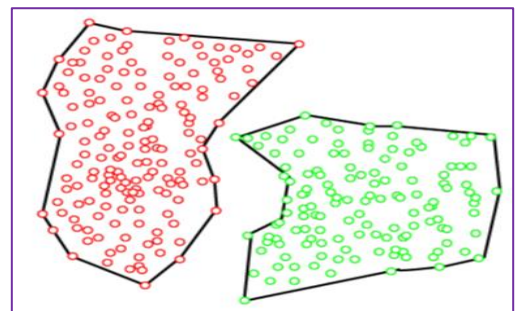
1. Partitioning Clustering

It is a type of clustering that divides the data into non-hierarchical groups. It is also known as the **centroid-based method**. The most common example of partitioning clustering is the K-Means Clustering algorithm. In this type, the dataset is divided into a set of k groups, where k is used to define the number of pre-defined groups. The cluster center is created in such a way that the distance between the data points of one cluster is minimum as compared to another cluster centroid.



2. Density-Based Clustering

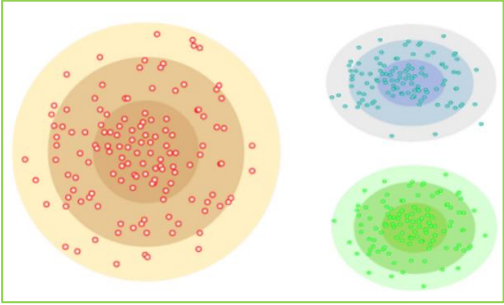
The density-based clustering method connects the highly-dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected. This algorithm does it by identifying different clusters in the dataset and connects the areas of high densities into clusters. The dense areas in data space are divided from each other by sparser areas. These algorithms can face difficulty in clustering the data points if the dataset has varying densities and high dimensions.



3. Distribution Model-Based Clustering

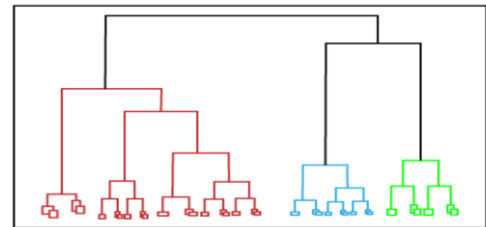
In the distribution model-based clustering method, the data is divided based on the probability of how a dataset belongs to a particular distribution. The grouping is done by assuming some distributions commonly **Gaussian Distribution**.

The example of this type is the **Expectation-Maximization Clustering algorithm** that uses Gaussian Mixture Models (GMM).



4. Hierarchical Clustering

Hierarchical clustering can be used as an alternative for the partitioned clustering as there is no requirement of pre-specifying the number of clusters to be created. In this technique, the dataset is divided into clusters to create a tree-like structure, which is also called a **dendrogram**. The observations or any number of clusters can be selected by cutting the tree at the correct level. The most common example of this method is the **Agglomerative Hierarchical algorithm**.



K- Means clustering

K-Means Clustering is an unsupervised learning algorithm that is used to solve the clustering problems in machine learning or data science. The k-means algorithm is one of the most popular clustering algorithms. It classifies the dataset by dividing the samples into different clusters of equal variances. The number of clusters must be specified in this algorithm.

Steps involved K-Means Clustering:

The working of the K-Means algorithm is explained in the below steps:

- Select the number K to decide the number of clusters.
- Select random K points or centroids. (It can be other from the input dataset).
- Assign each data point to their closest centroid, which will form the predefined K clusters.
- Calculate the variance and place a new centroid of each cluster.
- Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.
- If any reassignment occurs, then go to step-4 else go to FINISH.
- The model is ready.

Activity: Visual AI: This tool allows you to visualize K-means clustering in real-time. Upload your own data or use provided examples, adjust parameters, and see how clusters change visually using the link [Visualise k-means](#)

Applications of K-Means Clustering:

- Market Segmentation: group customers based on similar purchasing behaviours or demographics for tailored marketing strategies.
- Image Segmentation: partition images into regions of similar colours to aid in tasks like object detection and compression.
- Document Clustering: categorize documents based on content similarity, aiding in organization and information retrieval.
- Anomaly Detection: identify outliers by clustering normal data points and detecting deviations.
- Customer Segmentation: segment customers for targeted marketing and personalized experiences.

Advantages of K-Means Clustering:

- Easy to implement, making it suitable for users of all levels.
- Handles large datasets with low computational resources.
- Works well with numerous features and data points.
- Are easy to understand, aiding in decision-making.
- Applicable across various domains and data types.

Limitations of K-Means Clustering:

- Results can vary based on initial centroid placement.
- Assumes clusters are spherical, which is not always true.
- Number of clusters must be known beforehand.
- Outliers can distort clusters due to their influence on centroids.
- May converge to suboptimal solutions instead of the global optimum.

For advanced Learners – Python Program for K Means Clustering

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans

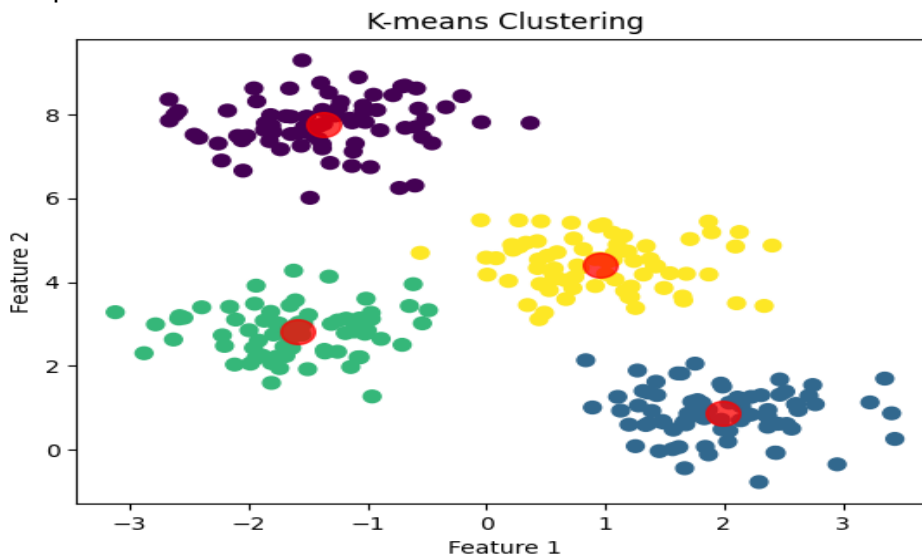
# Generate synthetic data
X, _ = make_blobs(n_samples=300, centers=4, cluster_std=0.60, random_state=0)

# Apply K-means clustering
kmeans = KMeans(n_clusters=4)
kmeans.fit(X)
y_kmeans = kmeans.predict(X)

# Plot the data points and centroids
plt.scatter(X[:, 0], X[:, 1], c=y_kmeans, s=50, cmap='viridis')
centers = kmeans.cluster_centers_
```

```
plt.scatter(centers[:, 0], centers[:, 1], c='red', s=200, alpha=0.75)
plt.title('K-means Clustering')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.show()
```

Output



This program does the following:

1. Generates synthetic data using `make_blobs` from `sklearn.datasets`.
2. Applies K-means clustering with `n_clusters=4`.
3. Plots the data points colored by their cluster assignments and shows the centroids as red circles.

You can adjust the parameters like the number of clusters, standard deviation, and number of samples in `make_blobs` to observe different clustering scenarios.

REFERENCES

Video Session

Videos: Clustering: <https://www.youtube.com/watch?v=wk2yII1qgU0>

K-means clustering: <https://www.youtube.com/watch?v=4b5d3muPQmA&t=119s>

EXERCISES

A. Multiple Choice Questions

1. Which of the following are the types of correlation?
 - a. Positive correlation
 - b. Negative Correlation
 - c. No correlation
 - d. All of the above
2. Which of the following techniques is an analysis of the relationship between two variables to provide the prediction mechanism?
 - a. Standard error
 - b. Correlation
 - c. Regression
 - d. None of the above

3. Which of the given plots is suitable for testing the linear relationship between a dependent and independent variable?
 - a. Bar chart
 - b. Scatter plot
 - c. Histograms
 - d. All of the above
4. Which of the following scatter plots represents a positive correlation?
 - a. points scattered randomly with no apparent trend
 - b. points forming a diagonal line and bottom left to top right
 - c. points forming a diagonal line from top left to bottom right
 - d. points clustered around a central point
5. Which regression technique is used when there is only one independent variable?
 - a. logistic regression
 - b. multiple linear regression
 - c. simple linear regression
 - d. polynomial regression
6. What is one advantage of linear regression analysis?
 - a. it is robust to outliers
 - b. it can capture nonlinear relationships between variables
 - c. it is simple and easy to interpret
 - d. it is suitable for classification tasks
7. What is supervised learning in Artificial Intelligence?
 - a. training a computer algorithm on input data that is not labelled.
 - b. training a computer algorithm on input data that has been labelled for a specific output.
 - c. training a computer algorithm without any input data
 - d. training a computer algorithm to perform unsupervised tasks.
8. Which type of classification involves categorizing data into two distinct classes?
 - a. multi-class classification
 - b. binary classification
 - c. unsupervised classification
 - d. regression classification
9. What is logistic regression commonly used for in binary classification?
 - a. categorizing observations into multiple classes
 - b. predicting continuous values for input data
 - c. categorizing observations into two distinct classes
 - d. identifying unstructured data patterns
10. What is the primary goal of classification in AI?
 - a. categorizing data into random groups
 - b. locating and classifying things or concepts into predefined groups
 - c. predicting continuous values for input data
 - d. identifying unstructured data patterns

11. Which algorithm is commonly used for binary classification?
 - a. Decision trees
 - b. Support Vector Machine
 - c. Logistic Regression
 - d. k-Nearest Neighbors
12. The K-Nearest Neighbors (KNN) algorithm assigns a class to new data point by considering:
 - a. Distance from the data point to a predefined decision boundary
 - b. Majority vote of its K nearest neighbors in the training data
 - c. Similarity of the data point to a cluster centroid
 - d. probability of each class given the data point's features.
13. What does a classification model in AI ultimately want to achieve?
 - a. to identify patterns and associations in data
 - b. to predict continuous numerical values
 - c. to categorize input data into predefined classes or labels
 - d. to optimize decision-making processes
14. What are some challenges in applying classification models to real-world problems?
 - a. Data bias and fairness
 - b. Interpretability and explainability
 - c. overfitting and underfitting
 - d. All of the above
15. What is clustering?
 - a. Grouping labeled dataset
 - b. Dividing data into different clusters
 - c. Finding linear association between variables
 - d. Predicting future behaviors of a dependent variable
16. Which type of learning does clustering belong to?
 - a. Supervised learning
 - b. Unsupervised learning
 - c. Semi-supervised learning
 - d. Reinforcement learning
17. Which method is used to group highly dense areas into clusters?
 - a. Partitioning clustering
 - b. Density-based clustering
 - c. Distribution model-based clustering
 - d. Hierarchical clustering
18. Which algorithm is an example of partitioning clustering?
 - a. Mean-shift algorithm
 - b. DBSCAN algorithm
 - c. K-Means algorithm
 - d. Fuzzy clustering algorithm

19. Which clustering method allows data objects to belong to more than one group or cluster?

- a. Partitioning clustering
- b. Density-based clustering
- c. Distribution model-based clustering
- d. Fuzzy clustering

20. Which clustering algorithm is sensitive to outliers?

- a. K-Means algorithm
- b. Mean-shift algorithm
- c. DBSCAN algorithm
- d. Hierarchical clustering

B. Fill in the blanks

1. In _____ type of ML, the models are not trained in labeled data sets.
2. The _____ measures the linear relationship between the independent and dependent variables.
3. _____ predicts _____ continuous _____ numerical values, while Logistic regression predicts discrete categories.
4. _____ are data points on the scatterplot that do not follow the pattern of the dataset.
5. _____ algorithm operates based on the principle of proximity, making predictions by considering the similarity between data points.
6. Clustering is a machine learning technique used to group _____ dataset.
7. Partitioning clustering divides the data into non-hierarchical groups, also known as _____ method.
8. Density-based clustering connects highly dense areas into clusters, separated by areas of _____.
9. The primary requirement for the number of clusters in K-Means algorithm is _____ beforehand.
10. Clustering is widely used in applications such as market segmentation and _____.

C. True or False:

1. Clustering is a supervised learning technique.
2. Hierarchical Clustering requires pre-specifying the number of clusters.
3. Fuzzy clustering is a hard clustering method.
4. Classification is an unsupervised learning technique.
5. In k-NN algorithm, k is the number of nearest data points.
6. K-Means algorithm requires specifying the number of clusters.

D. Short answer type questions:

1. What is Machine learning? Name the three methods of machine learning.
2. How are correlation measures used in AI applications?
3. Name some examples of regression algorithms?
4. What are regression algorithms used for?
5. What is Linear regression? Give two applications of regression in machine learning?
6. How can outliers impact regression analysis?
7. What is the primary difference between classification and regression?
8. Provide examples of classification problems in real-life scenarios.
9. What are some common applications of clustering techniques?
10. List the types of clustering methods.

E. Long Answer Questions:

1. How does classification model work?
2. Explain the types of clustering.
3. Write any two advantages and disadvantages of linear regression.
4. What are the steps involved in k-NN algorithm?
5. What are the steps involved in k-means clustering?